CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. 2.8×10^{-3} mol of CO_2 is left after removing 10^{21} molecules from its 'x' mg sample. The mass of CO_2 taken initially is

Given : $N_A = 6.02 \times 10^{23}$ mol^{-1}

(1) 98.3 mg (2) 48.2 mg

(3) 150.4 mg (4) 196.2 mg

Answer (4)

Sol. Moles of removed $CO_2 = \frac{10^{21}}{6.02 \times 10^{23}}$ mol

$= 1.66 \times 10^{-3}$ mol

mole of CO_2 left = 2.8×10^{-3} moles

total moles of CO_2 taken initially

$= (2.8 + 1.66) \times 10^{-3}$ mol

mass of CO_2 taken initially

$= 4.46 \times 10^{-3} \times 44$

$= 196.24 \times 10^{-3}$ g

= 196.24 mg

52. $FeO_4^{2-} \xrightarrow{+2.0\,V} Fe^{3+} \xrightarrow{0.8\,V} Fe^{2+} \xrightarrow{-0.5\,V} Fe^0$

In the above diagram, the standard electrode potentials are given in volts (over the arrow).

The value of $E^{\ominus}_{FeO_4^{2-}/Fe^{2+}}$ is

(1) 1.7 V (2) 2.1 V

(3) 1.4 V (4) 1.2 V

Answer (1)

Sol. $FeO_4^{2-} + 3e^- \rightarrow Fe^{3+}$; $E^\circ = 2.0$ eV

$Fe^{3+} + e^- \rightarrow Fe^{2+}$; $E^\circ = 0.8$ eV

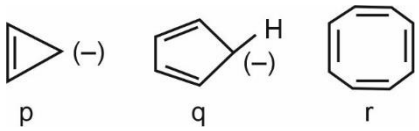
$FeO_4^{2-} + 4e^- \rightarrow Fe^{2+}$; $E^\circ = x$ (say)

$4 \times x = 3 \times 2 + 1 \times 0.8$

$x = \frac{6.8}{4}$ V

$= 1.7$ V

53. The correct stability order of the following species/molecules is



p q r

(1) q > p > r (2) q > r > p

(3) r > q > p (4) p > q > r

Answer (2)

Sol. q is aromatic

r is non-aromatic

p is antiaromatic

q > r > p (order of stability)

Aromatic > non-aromatic > antiaromatic

54. Ice at –5ºC is heated to become vapour with temperature of 110ºC at atmospheric pressure. The entropy change associated with this process can be obtained from

(1) $\int_{268K}^{273K} C_{p,m}dT + \frac{\Delta H_{m,fusion}}{T_f} + \frac{\Delta H_{m,vaporisation}}{T_b} + \int_{273K}^{373K} C_{p,m}dT + \int_{373K}^{383K} C_{p,m}dT$

(2) $\int_{268K}^{383K} C_p\,dT + \frac{\Delta H_{melting}}{273} + \frac{\Delta H_{boiling}}{373}$

(3) $\int_{268K}^{273K} \frac{C_{p,m}}{T}dT + \frac{\Delta H_{m,fusion}}{T_f} + \frac{\Delta H_{m,vaporisation}}{T_b} + \int_{273K}^{373K} \frac{C_{p,m}dT}{T} + \int_{373K}^{383K} \frac{C_{p,m}dT}{T}$

(4) $\int_{268K}^{383K} C_p dT + \frac{q_{rev}}{T}$

Answer (3)



Sol. $\underset{-5^\circ C}{H_2O(s)} \rightarrow \underset{0^\circ C}{H_2O(s)}$; $\Delta S_1 = \int_{268K}^{273K} \frac{C_{p,\,m}dT}{T}$

$\underset{0^\circ C}{H_2O(s)} \rightleftharpoons \underset{0^\circ C}{H_2O(l)}$; $\Delta S_2 = \frac{\Delta H_m,\, fus}{273}$

$\underset{0^\circ C}{H_2O(l)} \rightarrow \underset{100^\circ C}{H_2O(l)}$; $\Delta S_3 = \int_{273}^{373} \frac{C_{p,\,m}dt}{T}$

$\underset{100^\circ C}{H_2O(l)} \rightleftharpoons \underset{100^\circ C}{H_2O(g)}$; $\Delta S_4 = \frac{\Delta H_m,\, vap}{373}$

$\underset{100^\circ C}{H_2O(g)} \rightarrow \underset{110^\circ C}{H_2O(g)}$; $\Delta S_5 = \int_{373}^{383} \frac{C_{p,\,m}dT}{T}$

$\Delta S_{total} = \Delta S_1 + \Delta S_2 + \Delta S_3 + \Delta S_4 + \Delta S_5$

55. Heat treatment of muscular pain involves radiation of wavelength of about 900 nm. Which spectral line of H atom is suitable for this?

Given: Rydberg constant $R_H = 10^5\ cm^{-1}$, $h = 6.6 \times 10^{-34}$ J s, $c = 3 \times 10^8$ m/s)

(1) Lyman series, $\infty \rightarrow 1$

(2) Paschen series, $5 \rightarrow 3$

(3) Paschen series, $\infty \rightarrow 3$

(4) Balmer series, $\infty \rightarrow 2$

Answer (3)

Sol. $\frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

$\frac{1}{\lambda} = 10^5 \left(\frac{1}{3^2} - \frac{1}{\infty^2}\right)$

$\frac{1}{\lambda} = 10^5 \times \frac{1}{9}$

$\lambda = 9 \times 10^{-5}$

$\lambda = 900 \times 10^{-7}$cm

$= 900$ Å

56. The major product of the following reaction is

$$CH_3CH_2CH = O \xrightarrow[\text{reflux}]{\text{excess HCHO alkali}} ?$$

(1) $CH_3 - CH_2 - CH_2 - OH$

(2) $CH_3 - \underset{\displaystyle CH_2 - OH}{\overset{\displaystyle |}{\underset{\displaystyle |}{CH}}} - CH = O$

(3) $CH_3 - \underset{\displaystyle CH_2 - OH}{\overset{\displaystyle CH_2 - OH}{\overset{\displaystyle |}{\underset{\displaystyle |}{C}}}} - CH_2 - OH$

(4) $CH_3 - \underset{\displaystyle CH_2}{\overset{\displaystyle O}{\overset{\displaystyle ||}{\underset{\displaystyle |}{C}}}} - CH = O$

Answer (3)

Sol. Propanal undergoes aldol condensation with excess of HCHO in presence of OH^- ions to 2,2-dihydroxy methyl propanal which further reacts with HCHO and undergoes Cannizzaro reaction to give 2,2-hydroxymethylpropan-1-ol.

$$CH_3 - CH_2 - \underset{\displaystyle O}{\overset{\displaystyle ||}{C}} - H + 2HCHO \xrightarrow{OH^-} \underset{\displaystyle CH_2 - OH}{\overset{\displaystyle CH_2OH}{\overset{\displaystyle |}{\underset{\displaystyle |}{CH_3 - C - CHO}}}}$$

$$\underset{\displaystyle CH_2 - OH}{\overset{\displaystyle CH_2 - OH}{\overset{\displaystyle |}{\underset{\displaystyle |}{CH_3 - C - CHO}}}} + HCHO \xrightarrow[\text{reflux}]{OH^-} \underset{\displaystyle CH_2 - OH}{\overset{\displaystyle CH_2 - OH}{\overset{\displaystyle |}{\underset{\displaystyle |}{CH_3 - C - CH_2OH}}}} + HCOO^-$$



57. Match the **LIST-I** with **LIST-II**.

	LIST-I (Classification of molecules based on octet rule)		**LIST-II (Example)**
A.	Molecules obeying octet rule	I.	NO, NO_2
B.	Molecules with incomplete octet	II.	BCl_3, $AlCl_3$
C.	Molecules with incomplete octet with odd electron	III.	H_2SO_4, PCl_5
D.	Molecules with expanded octet	IV.	CCl_4, CO_2

Choose the ***correct*** answer from the options given below.

(1) A-IV, B-I, C-III, D-II (2) A-II, B-IV, C-III, D-I

(3) A-IV, B-II, C-I, D-III (4) A-III, B-II, C-I, D-IV

Answer (3)

Sol. $NO = 7e^-$, $NO_2 = 7e^-$ } Incomplete octet with odd electron

$BCl_3 = 6e^-$, $AlCl_3 = 6e^-$ } Incomplete octet

$H_2SO_4 = 12e^-$, $PCl_5 = 10e^- \Rightarrow$ molecules with expanded octet

$CCl_4 = 8e^-$, $CO_2 = 8e^- \Rightarrow$ molecules obeying octet rule

58. Math the **LIST-I** with **LIST-II**.

	LIST-I (Name reaction)		**LIST-II (Product obtainable)**
A.	Swarts reaction	I.	Ethyl benzene
B.	Sandmeyer's reaction	II.	Ethyl iodide
C.	Wurtz-Fittig reaction	III.	Cyanobenzene
D.	Finkelstein reaction	IV.	Ethyl fluoride

Choose the ***correct*** answer from the options given below.

(1) A-II, B-I, C-III, D-IV (2) A-IV, B-I, C-III, D-II

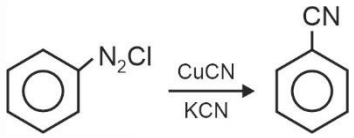
(3) A-IV, B-III, C-I, D-II (4) A-II, B-III, C-I, D-IV

Answer (3)

Sol. Swarts rection : Halogen exchange rxn

$$CH_3 - Br + AgF \xrightarrow{DMF} CH_3F$$

Sandmeyer's reaction :



Wurtz-Fittig reaction :

$$Ph - Br + C_2H_5Br \xrightarrow[\text{Dry ether}]{Na} Ph - C_2H_5$$

Finkelstein reaction : Halogen exchange rxn

$$R - Cl \xrightarrow[\text{Acetone}]{NaI} R - I$$

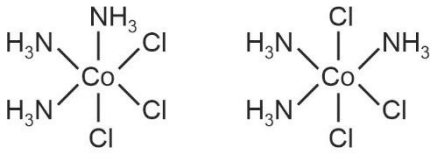
59. The complex that shows Facial - Meridional isomerism is

(1) $[Co(en)_2Cl_2]^+$ (2) $[Co(en)_3]^{3+}$

(3) $[Co(NH_3)_4Cl_2]^+$ (4) $[Co(NH_3)_3Cl_3]$

Answer (4)

Sol. Ma_3b_3 type of complex can show Facial-Meridional type isomerism.(where a and b are monodentate ligands)





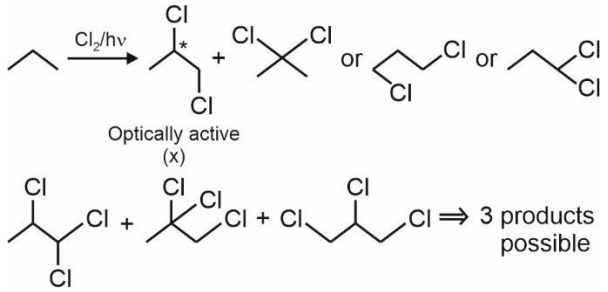
60. Propane molecule on chlorination under photochemical condition gives two di-chloro products, "x" and "y". Amongst "x" and "y", "x" is an optically active molecule. How many tri-chloro products (consider only structural isomers) will be obtained from "x' when it is further treated with chlorine under the photochemical condition?

(1) 4 (2) 3

(3) 2 (4) 5

Answer (2)

Sol.



61. The correct set of ions (aqueous solution) with same colour from the following is:

(1) Ti^{4+}, V^{4+}, Mn^{2+} (2) Zn^{2+}, V^{3+}, Fe^{3+}

(3) V^{2+}, Cr^{3+}, Mn^{3+} (4) Sc^{3+}, Ti^{3+}, Cr^{2+}

Answer (3)

Sol. V^{2+} = Violet

Cr^{3+} = Violet

Mn^{3+} = Violet

62. $CrCl_3.xNH_3$ can exist as a complex. 0.1 molal aqueous solution of this complex shows a depression in freezing point of 0.558°C. Assuming 100% ionisation of this complex and coordination number of Cr is 6, the complex will be

(Given K_f = 1.86 K kg mol^{-1})

(1) $[Cr(NH_3)_4Cl_2]$ Cl (2) $[Cr(NH_3)_3Cl_3]$

(3) $[Cr(NH_3)_6]$ Cl_3 (4) $[Cr(NH_3)_5Cl]$ Cl_2

Answer (4)

Sol. $\Delta T_f = iK_fm$

$0.558 = i \times 1.86 \times 0.1$

$$i = \frac{0.558}{0.186} = 3$$

Number of ions when 100% ionisation takes place = 3

$$[Cr(NH_3)_5Cl]Cl_2 \xrightarrow{\text{aqueous solution}} [Cr(NH_3)_5Cl]^{2+} + 2Cl^-$$

Number of ions = 3

63. What amount of bromine will be required to convert 2 g of phenol into 2,4,6-tribromophenol?

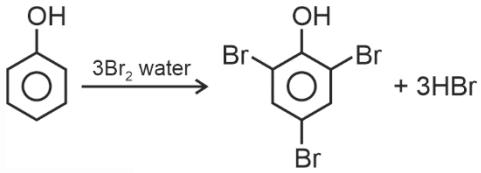
(Given molar mass in g mol^{-1} of C, H, O, Br are 12, 1, 16, 80 respectively)

(1) 4.0 g (2) 6.0 g

(3) 10.22 g (4) 20.44 g

Answer (3)

Sol.



Moles of phenol = $\frac{2}{94}$ mol

Moles of $Br_2 = \frac{3 \times 2}{94}$ moles

Mass of $Br_2 = \frac{3 \times 2}{94} \times 160$ g

= 10.22 g

64. Given below are two statements:

Statement I: Fructose does not contain an aldehydic group but still reduces Tollen's reagent.

Statement II: In the presence of base, fructose undergoes rearrangement to give glucose.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both Statement I and Statement II are false

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are true

Answer (4)



Sol. Fructose has α-hydroxy ketone group which tautomerise to aldehyde group in presence of base, therefore it reduces Tollen's reagent.

In presence of base, fructose undergo rearrangement to give glucose and mannose.

65. The d-electronic configuration of an octahedral Co(II) complex having magnetic moment of 3.95 BM is:

(1) $t_{2g}^5 e_g^2$ (2) $t_{2g}^3 e_g^0$

(3) $t_{2g}^6 e_g^1$ (4) $e^4 t_2^3$

Answer (1)

Sol. Co^{2+} complex having μ = 3.95 BM

Hence number of unpaired electron = 3

$Co^{2+} \Rightarrow 3d^7 = t_{2g}^5 e_g^2$

66. Which of the following happens when NH_4OH is added gradually to the solution containing 1 M A^{2+} and 1 M B^{3+} ions?

Given : $K_{sp}[A(OH)_2] = 9 \times 10^{-10}$ and $K_{sp}[B(OH)_3] = 27 \times 10^{-18}$ at 298 K

(1) $B(OH)_3$ will precipitate before $A(OH)_2$

(2) $A(OH)_2$ will precipitate before $B(OH)_3$

(3) $A(OH)_2$ and $B(OH)_3$ will precipitate together

(4) Both $A(OH)_2$ and $B(OH)_3$ do not show precipitation with NH_4OH

Answer (1)

Sol. Condition for precipitation IP > K_{sp}

For, $[A(OH)_2]$

$[A^{2+}][OH^-]^2 > 9 \times 10^{-10}$

$[A^{2+}] = 1$ M

$[OH^-] > 3 \times 10^{-5}$

For, $[B(OH)_3]$

$[B^{3+}][OH^-]^3 > 27 \times 10^{-18}$

$[B^{3+}] = 1$ M

$[OH^-] > 3 \times 10^{-6}$

So, $B(OH)_3$ will precipitate first

67. The **incorrect** statement among the following is

(1) PF_3 exists but NF_5 does not

(2) PH_3 shows lower proton affinity than NH_3

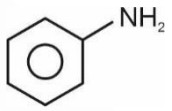
(3) SO_2 can act as an oxidizing agent, but not as a reducing agent

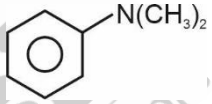
(4) NO_2 can dimerise easily

Answer (3)

Sol. $\overset{+4}{S}O_2$ can acts as both oxidising agent as well reducing agents because due to intermediate oxidation state, it can oxidise and reduce as well.

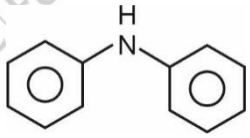
68. Which among the following react with Hinsberg's reagent?

A. 

B. 

C. $CH_3 — NH_2$

D. $N(CH_3)_3$

E. 

Choose the correct answer from the options given below

(1) A, B and E Only

(2) C and D Only

(3) A,C and E Only

(4) B and D Only

Answer (3)

Sol. Primary and secondary amines can reacts with Hinsberg reagent.



69. Given below are two statements

Statement I : In Lassaigne's test, the covalent organic molecules are transformed into ionic compounds.

Statement II: The sodium fusion extract of an organic compound having N and S gives prussian blue colour with $FeSO_4$ and $Na_4[Fe(CN)_6]$

In the light of the above statements, choose the correct answer from the options given below

(1) Both Statement I and Statement II are false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Statement I is true but Statement II is false

Answer (4)

Sol. Lassaigne's test is a general test for detection of halogen, nitrogen and sulphur in an organic compound. These elements covalently bonded to the organic compounds, In order to detect them, these have to converted into ionic forms.

$3Na_4[Fe(CN)_6] + 2Fe_2(SO_4)_3 \longrightarrow Fe_4[Fe(CN)_6]_3 + 6\ Na_2SO_4$ (prussian Blue)

70. The element that does not belong to the same period of the remaining elements (modern periodic table) is

(1) Iridium (2) Osmium

(3) Palladium (4) Platinum

Answer (3)

Sol. Ir, Os and Pt belong to 6th period of periodic table while Pd belongs to 5th period.

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

71. For the thermal decomposition of $N_2O_5(g)$ at constant volume, the following table can be formed, for the reaction mentioned below.

$2N_2O_5(g) \rightarrow 2N_2O_4(g) + O_2(g)$

Sr. No.	Time/s	Total pressure/(atm
1	0	0.6
2	100	'x'

$x = ___ \times 10^{-3}$ atm [nearest integer]

Given : Rate constant for the reaction is $4.606 \times 10^{-2}\ s^{-1}$.

Answer (897)

Sol. $2N_2O_5(g) \rightarrow 2N_2O_4(g) + O_2(g)$

$$k = \frac{2.303}{t}\log\frac{0.9-0.6}{(0.9-x)}$$

$$2 \times 10^{-2} \times 100 = \log\frac{0.3}{(0.9-x)}$$

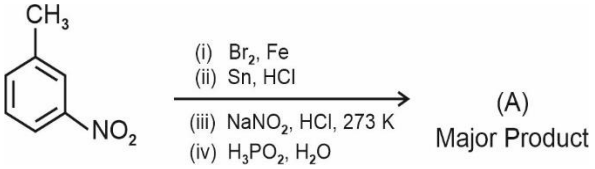
$$100 = \frac{0.3}{(0.9-x)}$$

$$= \frac{0.9-x}{0.3} = 0.01$$

$$0.9 - X = 0.003$$

$$= 897 \times 10^{-3}$$

72. Consider the following sequence of reactions to produce major product (A)



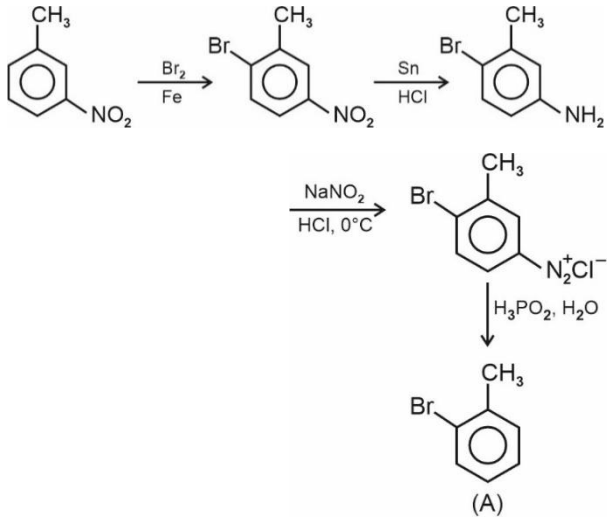
Molar mass of product (A) is ____g mol^{-1}.

(Given molar mass in g mol^{-1} of C : 12, H : 1, O : 16, Br : 80, N : 14, P : 31)



Answer (171)

Sol.



(A)

Molar mass of A = 171 g/mol

73. During "S" estimation, 160 mg of an organic compound gives 466 mg of barium sulphate. The percentage of Sulphur in the given compound is ____%.

(Given molar mass in g mol^{-1} of Ba : 137, S : 32, O : 16)

Answer (40)

Sol. m mole of $BaSO_4$ = mmoles of $S = \frac{466}{233}$

Mass of $S = \frac{466}{233} \times 32$ mg

= 64 mg

$\% \ S = \frac{64}{160} \times 100 = 40\%$

74. The standard enthalpy and standard entropy of decomposition of N_2O_4 to NO_2 are 55.0 kJ mol^{-1} and 175.0 J/K/mol respectively. The standard free energy change for this reaction at 25°C in J mol^{-1} is _____ (Nearest integer)

Answer (2850)

Sol. $\Delta H^\circ_{rxn} = 55.0$ kJ / mol T = 298 K

$\Delta S^\circ_{rxn} = 175$ J / mol

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$= 55000 - 298 \times 175$

= 2850 J/mol

75. If 1 mM solution of ethylamine produces pH = 9, then the ionization constant (K_b) of ethylamine is 10^{-x}. The value of x is _____ (nearest integer).

[The degree of ionization of ethylamine can be neglected with respect to unity.]

Answer (7)

Sol. pOH + pH = 14

pOH = 14–9

pOH = 5

$[OH^-] = 10^{-5}$ m

$[OH^-] = \sqrt{K_b \cdot C}$

C = concentration of weak base = 1 mM = 10^{-3} M

$10^{-5} = \sqrt{K_b \times 10^{-3}}$

$10^{-10} = K_b \times 10^{-3}$

$K_b = 10^{-7}$

x = 7

❑ ❑ ❑

